

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain Bone structure & composition & its composition.
- Q.24 Write a note on Hard Tissue replacement implants : Orthopaedics & Dental Implants.
- Q.25 Explain Effect of sterilization on properties of biomaterials.

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No. of Printed Pages : 4

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Roll No.

3rd Sem / Medical Electronics

Subject : Bio-Mechanics and Bio Materials

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Biomaterial may be used to
- a) Replace a non-functional organ
 - b) Augment any tissue function
 - c) Evaluate function of an organ
 - d) All of the above
- Q.2 Which of the following is not an example of biomaterial application.
- a) Cardiovascular stent
 - b) Heart valve
 - c) Eye glasses
 - d) Dental implant
- Q.3 Which is not a characteristic of biocompatible material
- a) good appearance
 - b) non toxic biodegradation
 - c) cytocompatible
 - d) non immunogenic

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- Q.4 The locations where biomaterials come in contact with cells are called
- Integrins
 - Focal adhesions
 - Surface interactors
 - Peptides
- Q.5 The following technique may not promote cellular attachment to biomaterials
- Increasing hydrophilicity of the surface
 - Keeping the surface ultra smooth
 - Creating a nanotopography on the surface
 - Adhereing protein fragments on the surface
- Q.6 A biomaterial intended for use as a replacement of a blood vessel_____.
- Must be rigid and have rough surfaces to promote cellular in growth
 - Must be rigid and must not degrade over time
 - Should be designed such that it encourages coagulation of blood
 - Must be flexible and have a porous structure

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Hemocompatibility

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- Q.8 Expand BMD
- Q.9 Bone is a _____ tissue type.
- Q.10 Biomechanics.
- Q.11 Viscoelasticity
- Q.12 Vascular implant

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Write a note on medical application of blood rheology.
- Q.14 Explain Newton laws of Motion.
- Q.15 Explain polymers and their classification.
- Q.16 Explain mechanical properties of Bone.
- Q.17 Explain structure and functions of soft tissues.
- Q.18 Write a note on classification of joint.
- Q.19 Explain mechanical properties of blood vessels.
- Q.20 Discuss various application of bio-materials.
- Q.21 Write a note on classification of sterilization
- Q.22 Explain bio-compatibility in details.

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